

SQL

1) Recap

- User persona
- Data stories
- Data modelling
- Metrics & KPI
- Analysis using Excel
- Visualization, interpretation & statistics

Question:

If we can do pivot tables & analysis in Excel → why still SQL?

2) Why SQL? And not Excel

- When business expands → different states / countries / zones
 - South zone, East city, North
 - Europe, Asia, Africa
 - Example: Amazon
 - When company grows to **global scale** → **Excel is not enough**
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Problems in Excel (Why not Excel?)

1. Scale of Data

- Row limit ≈ ~1 million rows (practical limit)
 - Very large files won't open
 - Application crashes
 - ~1 GB file becomes heavy
 - Big data cannot be handled properly in Excel
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2. Data Wrangling (Preparing data for analysis/visualization)

- Need to combine multiple tables
- Use VLOOKUP
- Complex when there are many tables (100 tables etc.)

3. Security

- Important sheets can easily be stolen

4. Versioning

- Updated data cannot be easily tracked
- Multiple copies:
 - final_v1
 - final_v2
 - final_final
- Group of 5–10 people working → confusion

5. Data Quality

- No strict constraints in Excel
 - Example: Age → 175, Twenty Two
- Need manual checking

6. Relationships

- Hard to build relationships between tables
- VLOOKUP based on common column
- Data checking becomes manual work

7. Backups & Recovery

- If system crashes / damaged → data gone

8. Standardization (Data issues)

- Different formats
 - Dates (dd/mm/yy etc.)
 - Numbers / currency formats
 - Sum / Count formats inconsistent
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9. Automation

- Not as good as SQL
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Best Alternative → SQL (Structured Query Language)

Storage

- Uses **RDBMS (Relational Database Management System)**
- SQL is the language used to communicate with it

RDBMS Structure

- Tables
 - Rows & Columns
 - Follows constraints
- Relationships between tables
- Can view relations in one place

Storage & Scalability

- Large scale storage
- Scalable

SQL Language

- English-like syntax
Example:

SELECT *

FROM student

WHERE stdname = 'Joe';

SQL Advantages

Data Wrangling

- Queries are powerful & useful

Security

- Authentication & Authorization
- Access control levels
 - Some can view only
 - Some can edit

Centralized Location

- Anyone with permission accesses latest data

Data Quality

- Constraints maintain correctness

Backup & Recovery

- Secondary location backups

Automation

- UDF (User Defined Functions)
- Stored Procedures
 - Store query & run it daily